

Biofilm Removal from Granular Media Filters for Microbial Analysis

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Abstract

When studying biofilter microbial communities, it is crucial to capture sufficient biomass. Filter media type, age, and operational status at sample collection impact yield of intact cells and DNA.

This study evaluated modifications to commercial DNA extraction kit procedures to maximize DNA recovery, as well as reduced duration of low frequency sonication (40 kHz) to maximize recovery of viable cells for flow cytometry (FCM).



Background & Problem Identification

Accurate microbial characterization depends on collecting bacterial cells at sufficient concentrations without compromising sample integrity. Techniques such as flow cytometry require well-dispersed, intact cells, while molecular analyses (qPCR, sequencing, metagenomics, qSIP) rely on high-quality nucleic acids. However, incomplete biofilm removal or cell damage during detachment can bias results and limit downstream analyses. Therefore, improved biofilm removal methods are needed to maximize biomass recovery while preserving cell and nucleic acid integrity.

1. Intact Cells



RESEARCH AIM:

To identify the optimal sonication time to detach biofilms from GAC for downstream microbial analysis.

2. DNA



RESEARCH AIM:

To evaluate different DNA extraction methods for maximum recovery of DNA

Research Approach

Key parameters of focus for maximizing biomass recovery from filter media

- **Sonication duration at low frequency.**
- **DNA extraction kit selection** (QIAGEN DNeasy PowerBiofilm Kit, Zymo Research Quick-DNA Fecal/Soil Microbe MidiPrep, QIAGEN DNeasy PowerSoil Pro Kit, and QIAGEN DNeasy PowerMax Soil) **bead-beating modification** (as prescribed, reduce speed, reduce duration, or invert tube),
- and **media collection timing** with respect to backwashing

Goal

Methods

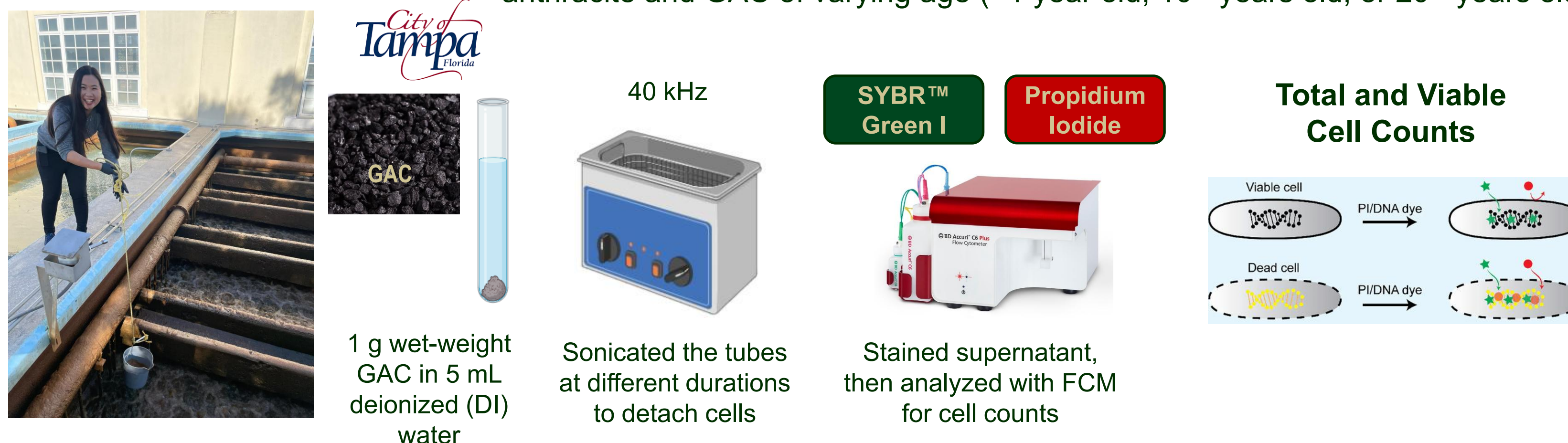
Biofilter Media

Table 1: Methods Overview

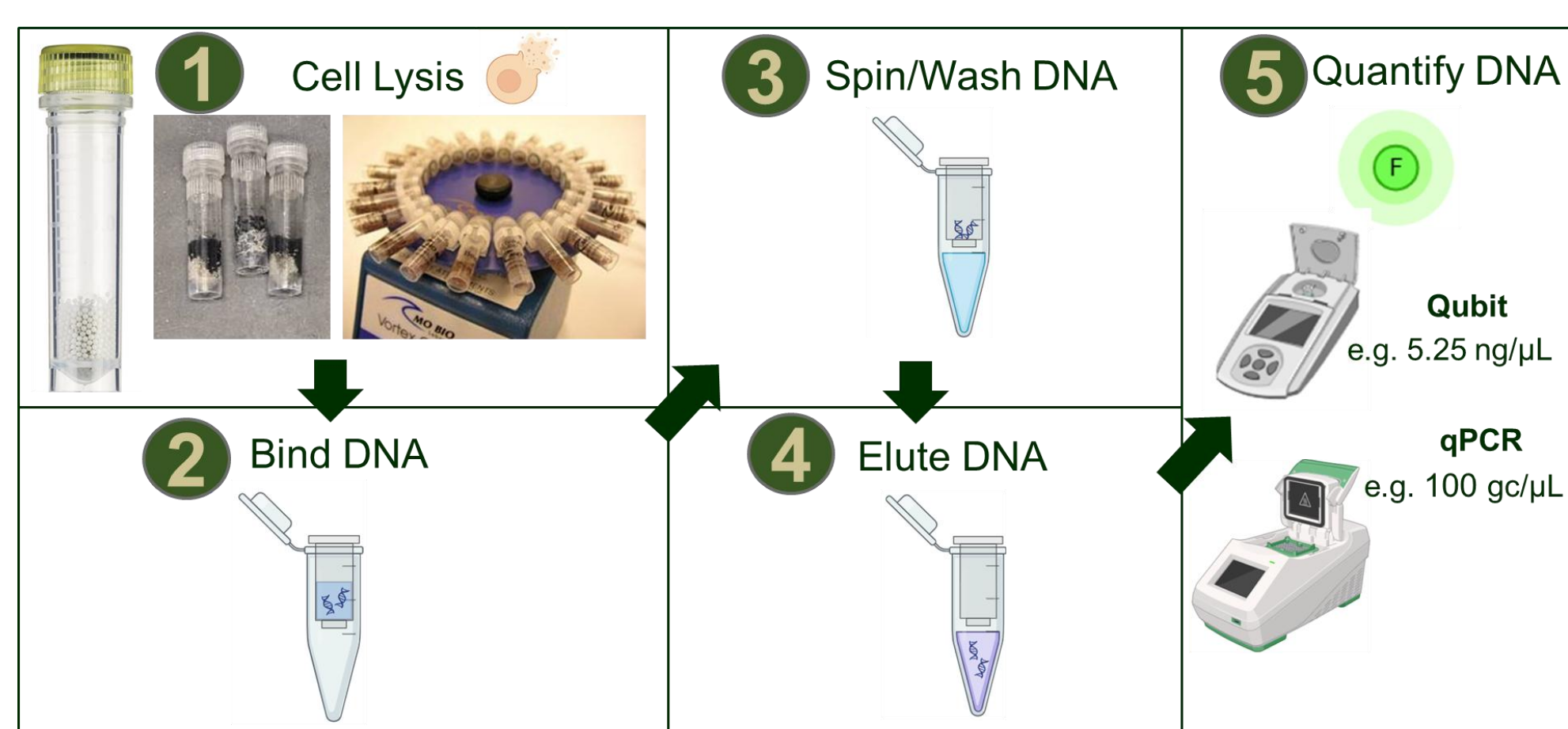
	Cells	DNA
	40 kHz	Extraction Kits
Sonication		
Biofilter Media		

Methods: Flow Cytometry

PURPOSE: to maximize intact cell and DNA yields. Biofilter media included anthracite and GAC of varying age (<1 year old, 10+ years old, or 20+ years old).



Methods: DNA Extraction for Molecular Applications



Reducing GAC breakage is crucial because GAC decomposition induced by sonication might interfere with subsequent analyses that rely on optical properties and/or reliable separation of target analytes.

Results & Discussion

Determining sonication time for FCM analysis

The highest intact and total cell counts were observed at sonication durations of 10 minutes (23.5 Ws/mL) and 15 minutes (35.2 Ws/mL), respectively (Table 3). Beyond 10-15 minutes, intact and total cell counts either plateaued or demonstrated considerable variability between samples, most likely due to the visually apparent degradation of GAC for durations ≥15 minutes

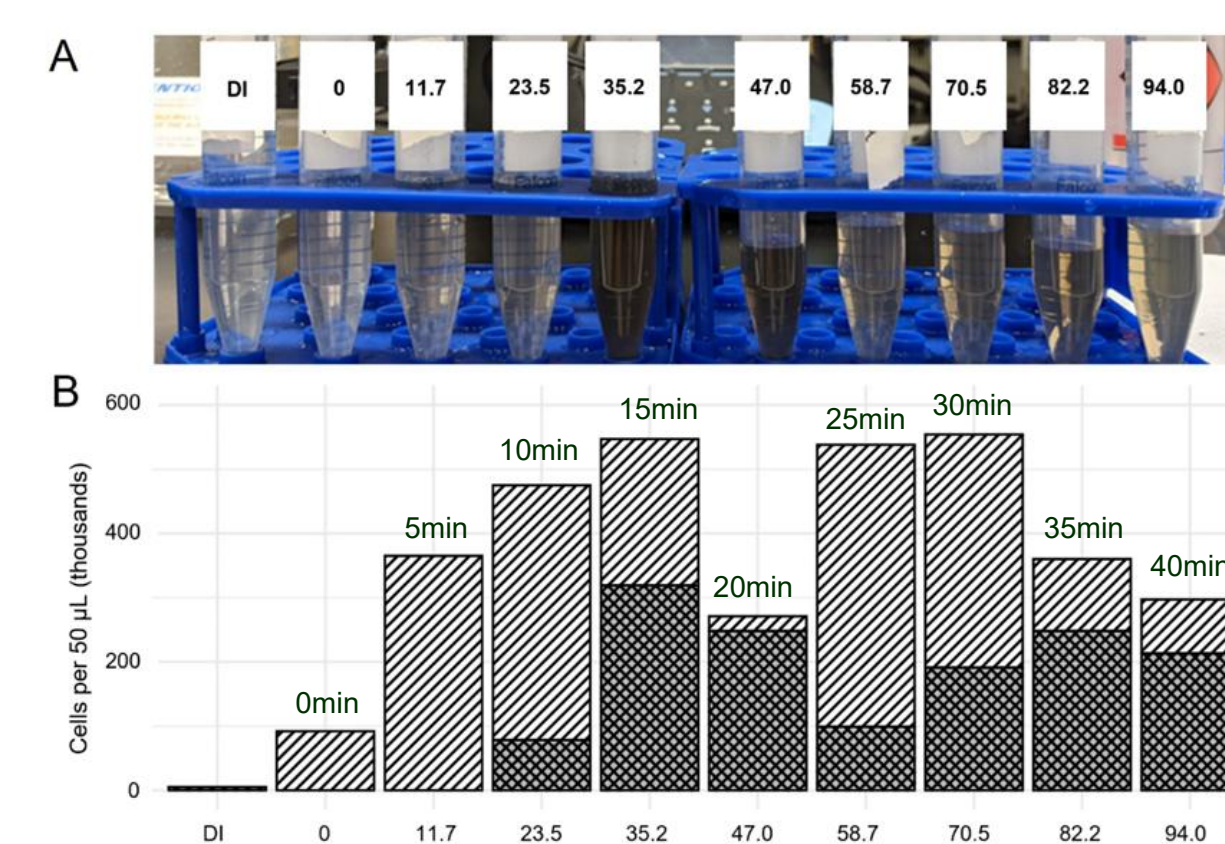


Figure 1. Intact and total (intact + lysed) cell counts via flow cytometry with intact/lysed cell staining for 'new' GAC media from filter F3 (1 year old). (A) visual sample quality and (B) cell counts are shown as a function of SED (0-40 min, corresponding to 0-94 Ws/mL). Deionized (DI) water is included as a background control.

Production of carbon fines during longer (≥15 minutes) sonication of GAC has also been observed in other work and was shown to increase linearly for a range of SEDs (0 to 400 Ws/mL), while recovery of DNA was strongly reduced at SEDs >67 Ws/mL (Kim et al., 2014).

Table 3: Total and intact cell results for each GAC age group, sonication duration, and corresponding sonication energy density (SED)

Filter Description	Sonication Duration Tested		SED Tested		Point of Maximum* Intact Cells		Point of Maximum Total Cells		Point of Sharp Decline in Total Cells	
	Increments (min)	Range (min)	Increments of SED (Ws/mL)	Range of SED (Ws/mL)	Duration at Maximum (min)	SED at Maximum (Ws/mL)	Duration at Maximum (min)	SED at Maximum (Ws/mL)	Duration at Minimum (min)	SED at Minimum (Ws/mL)
Newer GAC, F3	5	0-40	11.7	11.7-94.0	10	23.5	15	23.5	20	47.0
Newer GAC, F3	1	0-15	2.3	2.3-35.2	8	18.8	8	18.8	9	21.1
Older GAC, F1	1	0-15	2.3	2.3-35.2	8	18.8	8	18.8	10	23.5

*Point of maximum intact cells refers to the initial maximum of intact cells, or the point of diminishing return

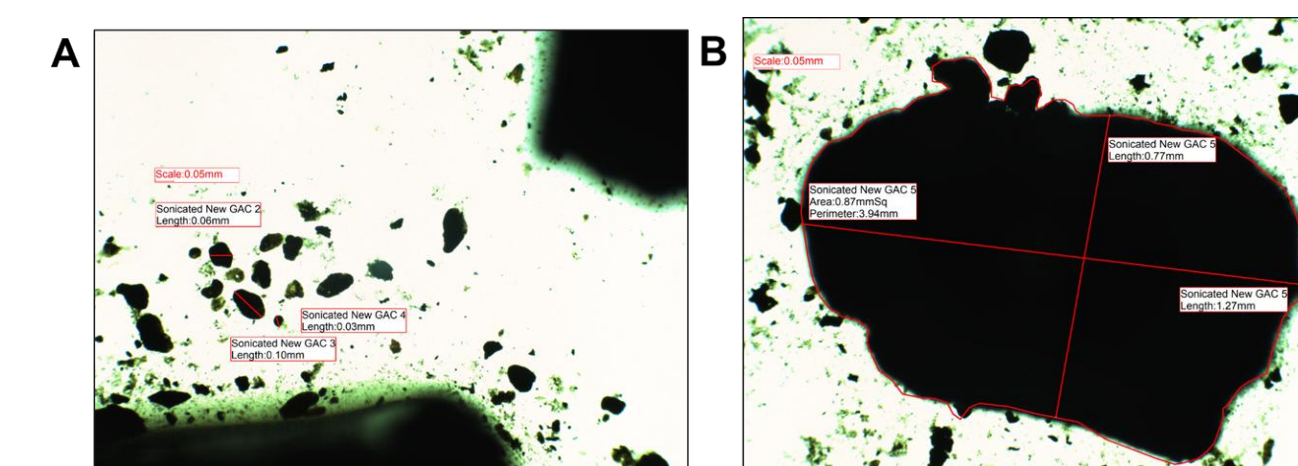


Figure 2 Micrographs of F3 GAC (< 1 year old) after nine minutes of sonication, showing representative grains 2-4 (A) and grain 5 (B) with dimensions shown in mm.

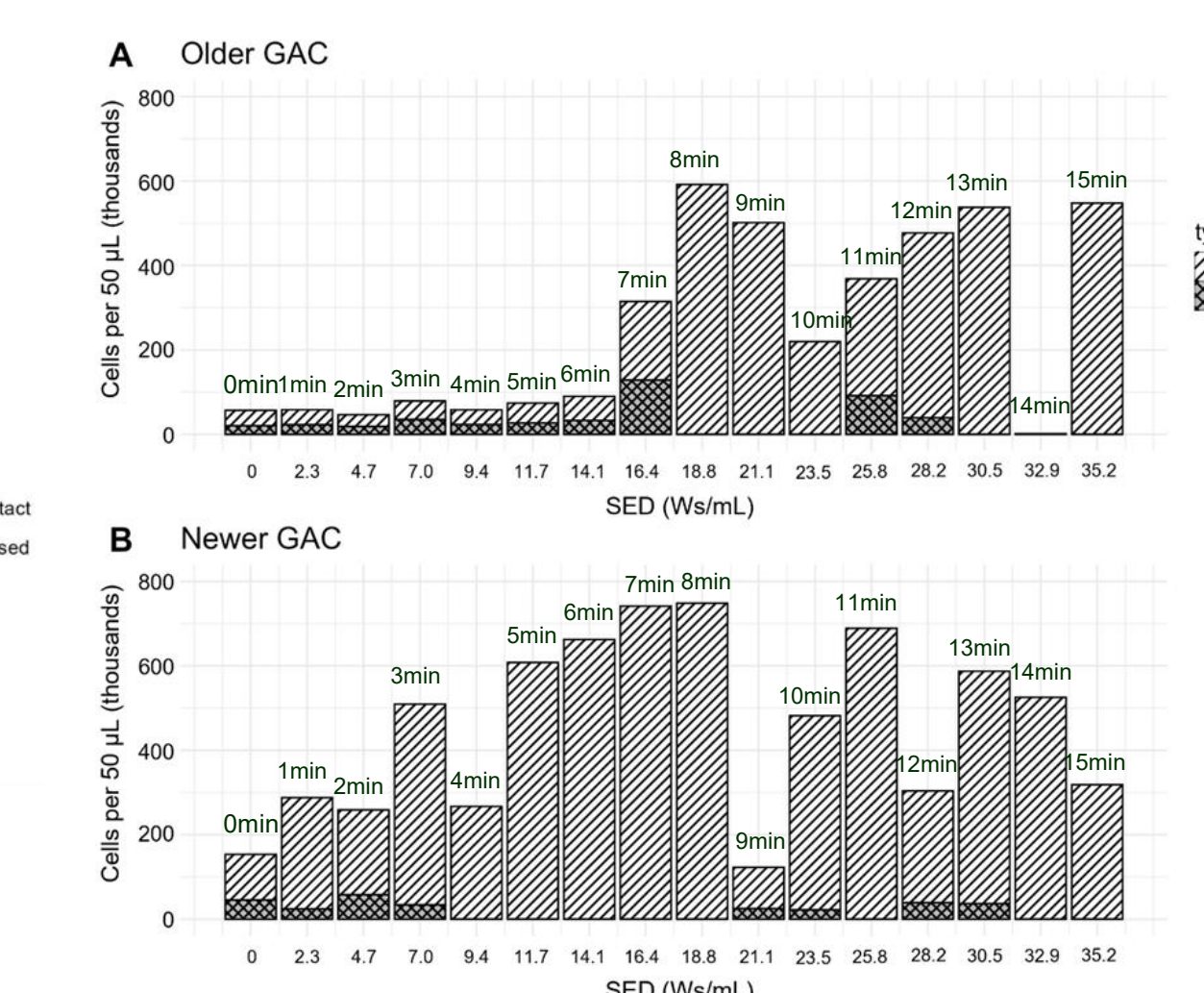


Figure 3. Comparison of intact cell counts per 50 μL of GAC eluate for different filter ages as a function of SED. (A) Older (20+ years old) GAC from filter F1 at 28 h runtime and (B) newer (1 year old) GAC from filter F3 at 26 h runtime. Duplicate media samples were analyzed for each minute of sonication, with bars representing the average of replicates.

$$SED = \frac{P_{GUT}}{V}$$

• SED : sonication energy density [Ws/mL]
• P_{GUT} : ultrasonic power delivered [W]
• t : duration for which power is applied [s]
• V : volume of water in the sonication

DNA extraction

Bead beating modifications:

1. Follow the kit (top speed, 10 min), ("none")
2. Vortex for less time, ("reduce duration")
3. Vortex at slower speed, ("reduce speed")
4. Invert the bead tube, ("invert")



Pro Kit ("soil pro"), and QIAGEN DNeasy PowerMax Soil ("soil max"). No bead-beating protocol is specified in the fecal/soil kit product guide.

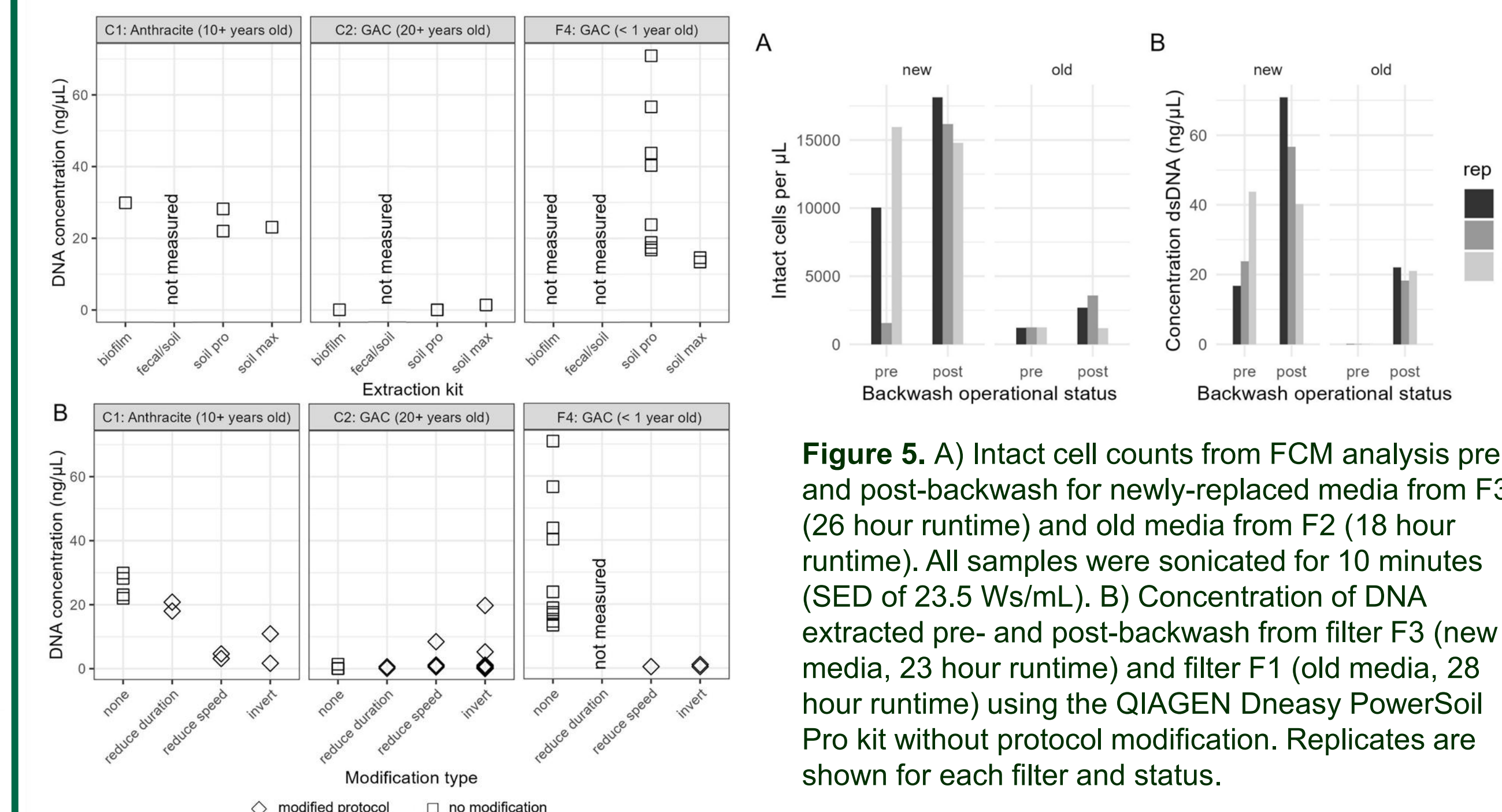


Figure 4. Resulting DNA concentrations when extracting from 1 gram (wet weight) of biofilter media using (A) various extraction kits without modification and (B) with various modifications. Squares represent results without modification to the kit protocol, while diamonds represent results of modifications. Kits tested included QIAGEN DNeasy PowerBiofilm Kit ("biofilm"), Zymo Research Quick-DNA Fecal/Soil Microbe MidiPrep ("fecal/soil"), QIAGEN DNeasy PowerSoil Pro Kit ("soil pro"), and QIAGEN DNeasy PowerMax Soil ("soil max"). No bead-beating protocol is specified in the fecal/soil kit product guide.

Result Discussion:

- GAC breakage (Figure 2) creates new adsorption sites for dissolved organic molecules such as DNA, thereby reducing extraction efficiency (Yapsakli et al., 2010).
- Carbon fines (Figure 2) are also known to reduce DNA extraction efficiency, and researchers speculate the main mechanism responsible is adsorption of DNA to carbon fines during extraction (Kim et al., 2014).
- Low-frequency sonication results in the breakdown of older carbon at extended sonication times. The lower DNA concentrations observed in older biofilter GAC could be a result of the carbon breaking apart during the bead-beating process and exposing internal pores in the GAC; future research should explore the exact mechanisms of GAC breakage and internal pore accessibility to determine the validity of this inference.

Conclusions & Ongoing Work



Harvesting biomass: Media type, age, and filter operational status are important

Useful in study design: When and where to collect media; sample site selection

Table 4: Summary of DNA extraction results for QIAGEN DNeasy PowerSoil Pro kit without modification. A runtime of 0 hours represents samples collected immediately after backwashing. Triplicate 0.25 g GAC samples were extracted for each filter and runtime. Data from filters F1 and F3 are also included in Figure 5B.

Filter description	Media Type	Runtime at Collection (hours) ^a	Media Age at Time of Study	[DNA] (ng/μL)
Water reuse column C1	Anthracite	0 (post)	10 + years	25.1 ± 3.1
Water reuse column C2	GAC	0 (post)	20 + years	0.043 ± 0.008
Drinking water F1	GAC	28 (pre)	20 + years	0.089 ± 0.005
Drinking water F1	GAC	0 (post)	20 + years	20.4 ± 1.6
Drinking water F3	GAC	23 (pre)	12 months	28.1 ± 11.4
Drinking water F3	GAC	0 (post)	12 months	56.0 ± 12.5

^aIndicates whether sample was collected pre-backwash or post-backwash

Key takeaway:

- Sonication duration exceeding 10 minutes (23.5 Ws/mL) at 40 kHz resulted in visual GAC breakage.
- Newer biofilter GAC yields higher intact cell counts and higher concentrations of DNA compared to old GAC.
- When working with older GAC, less-aggressive bead-beating (e.g., decreasing vortex speed and duration or avoiding vortexing altogether) improves DNA yields, and shorter sonication times (i.e., lower sonication energy densities) improve cell yields.
- Microbial DNA removal from anthracite biofilter media and from new GAC requires little modification to DNA extraction kits typically used in soil applications.

Ongoing Research: Investigating GAC Breakage, Biofilm Structure, and Microbial Community Shifts

- Explore the exact mechanisms of GAC breakage and internal pore accessibility
- Pair DNA extraction studies with microscopy to investigate biofilm thickness and coverage
- Characterize microbial communities in pre- and post-backwash samples

The results from both studies highlight the importance of considering media type, age, and filter runtime when conducting microbial analyses.

For more information on this research, please see the full publications:



Funding and Acknowledgements:

This research has received support from: the National Science Foundation under Grant No. 1832713. The authors would like to acknowledge the David L. Tippin Water Treatment Facility (City of Tampa Water Department) and the Clark County Water Reclamation District, as well as Dr. Jayme Peraud (USF) and Dr. Aarthi Mohan (SNWA).
References: Kim et al. (2014). J. Environ. Sci. Health A, 49, 609–616. Yapsakli et al. (2010). Process Biochem., 45, 1543–1549.